

TRUNCATIONS OF THE RING OF NUMBER-THEORETIC FUNCTIONS

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ABSTRACT. We study the ring Γ of all functions $\mathbb{N}^+ \rightarrow K$, endowed with the usual convolution product. Γ , which we call *the ring of number-theoretic functions*, is an inverse limit of the “truncations”

$$\Gamma_n = \{ f \in \Gamma \mid \forall m > n : f(m) = 0 \}.$$

Each Γ_n is a zero-dimensional, finitely generated K -algebra, which may be expressed as the quotient of a finitely generated polynomial ring with a *stable* (after reversing the order of the variables) monomial ideal. Using the description of the free minimal resolution of stable ideals given by Eliahou-Kervaire, and some additional arguments by Aramova-Herzog and Peeva, we give the Poincaré-Betti series for Γ_n .

1. INTRODUCTION

Cashwell and Everett [?] studied “the ring of number-theoretic functions”

$$\Gamma = \{ f \mid \mathbb{N}^+ \rightarrow K \} \quad (1)$$

where $\mathbb{N}^+$ is the set of positive natural numbers (we denote by $\mathbb{N}$ the set of all natural numbers) and K is a field containing the rational numbers. Γ is endowed with component-wise addition and multiplication with scalars, and with the convolution (or Cauchy) product

$$fg(n) = \sum_{\substack{(a,b) \in (\mathbb{N}^+) \times (\mathbb{N}^+) \\ ab=n}} f(a)g(b) \quad (2)$$

With these operations, Γ becomes a commutative K -algebra. It is immediate that it is a local domain; less obvious is the fact that it is a unique factorisation domain. Cashwell and Everett proved this in [?] using the isomorphism

$$\begin{aligned} \Phi : \Gamma &\rightarrow K[[X]] \\ f &\mapsto \sum f(n) x_1^{\alpha_1} x_2^{\alpha_2} \cdots \end{aligned} \quad (3)$$

where $X = \{x_1, x_2, x_3, \dots\}$, $K[[X]]$ is the “large” power series ring of all functions from the free abelian monoid $\mathcal{M} = [X]$ (the free abelian monoid generated by X) to K , and where the summation extends over all $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots \in \mathbb{N}^+$. Here, and henceforth, we denote by p_i the i ’th prime number, with $p_1 = 2$, and by $\mathcal{P}$ the set of all prime numbers. That (??) is an isomorphism is immediate from the following isomorphism of commutative monoids, implied by the fundamental theorem of arithmetics:

$$(\mathbb{N}^+, \cdot) \simeq \coprod_{p \in \mathcal{P}} (\mathbb{N}, +) \quad (4)$$

The following number-theoretic functions are of particular interest (whenever possible, we use the same notation as in [?]):

- (1) The multiplicative unit ϵ given by $\epsilon(1) = 1$, $\epsilon(n) = 0$ for $n > 1$,
- (2) $\lambda : \mathbb{N}^+ \rightarrow \mathbb{N}$ given by $\lambda(1) = 0$, $\lambda(q_1 \cdots q_l) = l$ if $q_1, \dots, q_l$ are any (not necessarily distinct) prime numbers.

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- (3) $\tilde{\lambda} : \mathbb{N}^+ \rightarrow \mathbb{N}$ given $\tilde{\lambda}(1) = 0$, $\tilde{\lambda}(p_1^{a_1} \dots p_r^{a_r}) = \sum a_r p_r$.
- (4) The Möbius function $\mu(1) = 1$, $\mu(n) = (-1)^v$ if n is the product of v distinct prime factors, and 0 otherwise,
- (5) For any $i \in \mathbb{N}^+$, $\chi_i(p_i) = 1$, and $\chi_i(m) = 0$ for $m \neq p_i$. Note that under the isomorphism $(??)$, $\Phi(\chi_i) = x_i$.

The topic of this article is the study of the “truncations” Γ_n , where for each $n \in \mathbb{N}^+$,

$$\Gamma_n = \{ f \in \Gamma \mid m > n \implies f(m) = 0 \} \quad (5)$$

With the modified multiplication given by

$$fg(n) = \sum_{\substack{(a,b) \in \{1, \dots, n\} \times \{1, \dots, n\} \\ ab=n}} f(a)g(b) \quad (6)$$

Γ_n becomes a K -algebra, isomorphic to Γ/J_n , where J_n is the ideal

$$J_n = \{ f \in \Gamma \mid \forall m \leq n : f(m) = 0 \}.$$

If we define

$$\pi_n : \Gamma \rightarrow \Gamma_n \quad (7)$$

$$\pi_n(f)(m) = \begin{cases} f(m) & m \leq n \\ 0 & m > n \end{cases} \quad (8)$$

then π_n is a K -algebra epimorphism, and J_n is the kernel of π_n . We note furthermore that J_n is generated by *monomials* in the elements χ_i .

To describe the main idea of this paper, we need a few additional definitions. First, for any $n \in \mathbb{N}^+$ we denote by $r(n) \in \mathbb{N}$ the largest integer such that $p_{r(n)} \leq n$. In other words, $r(n)$ is the number of prime numbers $\leq n$ (this number is often denoted $\pi(n)$). Secondly, for a monomial $m = x_1^{\alpha_1} \dots x_w^{\alpha_w}$, we define the *support* $\text{Supp}(m)$ as the set of positive integers i such that $\alpha_i > 0$. We define $\max(m)$ and $\min(m)$ as the maximal and minimal elements in the support of m .

Definition 1.1. A monomial ideal $I \subset K[x_1, \dots, x_r]$ is said to be *strongly stable* if whenever m is a monomial such that $x_j m \in I$, then $x_i m \in I$ for all $i \leq j$. If this condition holds at least for all $i \leq j = \max(m)$ then I is said to be *stable*.

We can now state our main theorem:

Theorem 1.2. *Let $n \in \mathbb{N}^+$ and $r = r(n)$. Then the following holds:*

- (I) $\Gamma_n \simeq \frac{K[x_1, \dots, x_r]}{I_n}$, where I_n is a strongly stable monomial ideal, with respect to the reverse order of the variables.
- (II) Γ_n is artinian, with $\dim_K(\Gamma_n) = n$. Furthermore, if it is given the natural grading with $|\chi_i| = 1$, then its Hilbert series is $\sum_i d_i t^i$ where d_i is the number of $w \leq n$ with $\lambda(w) = i$.
- (III) There is a 1-1 bijection between the minimal monomial generators of I_n of minimal support v , and the solutions in non-negative integers to the equation

$$\log n - \log p_v < \sum_{i=v}^r b_i \log p_i \leq \log n \quad (9)$$

- (IV) If we denote by $C_{n,v}$ the number of such solutions, then the Poincaré-Betti series of the free minimal resolution of K as a cyclic module over Γ_n is the following rational function:

$$P(\text{Tor}_*^{\Gamma_n}(K, K), t) = \frac{(1+t)^r}{1-t^2 \left(\sum_{i=1}^r (1+t)^{(i-1)} C_{n,r-i+1} \right)} \quad (10)$$

We will show this result, and also give the graded Poincaré-Betti series. For this, we define the number $C_{n,v,d}$ which counts the number of minimal generators of I_n of minimal support v and total degree d . We determine some elementary properties of the numbers $C_{n,v,d}$ and $C_{n,v}$.

2. THE RING OF NUMBER-THEORETIC FUNCTIONS AND ITS TRUNCATIONS

2.1. Norms, degrees, and multiplicativity. For a monomial $\mathcal{M} \ni m = x_1^{a_1} \dots x_n^{a_n}$ we define the *weight* of m as $w(m) = p_1^{a_1} \dots p_n^{a_n}$ (we put $w(1) = 1$). Hence w gives a bijection between $\mathcal{M}$ and $\mathbb{N}^+$. Furthermore, we can define a term order on $\mathcal{M}$ by $m > m'$ iff $w(m) > w(m')$. If we define the *initial monomial* $\text{in}(f)$ of $f \in K[[X]]$ as the monomial in $\text{Supp}(f)$ minimal with respect to $>$, then $\text{in}(f)$ is easily seen to correspond to the *norm* $N(\alpha)$ of a number-theoretic function α , defined as the smallest n such that $\alpha(n) \neq 0$. Here, we must use w and Φ to identify $\mathcal{M}$ and $\mathbb{N}^+$ and $K[[X]]$ and Γ . As observed in [?], the norm is multiplicative: $N(\alpha\beta) = N(\alpha)N(\beta)$.

Cashwell and Everett also define the *degree* $D(\alpha)$ to mean the smallest d such that there exists an n with $\lambda(n) = d$ and $\alpha(n) \neq 0$. This corresponds the smallest *total degree* of a monomial in $\text{Supp}(f)$. Furthermore, the norm $M(\alpha)$, defined as the smallest integer n with $\lambda(n) = D(\alpha)$, $\alpha(n) \neq 0$, corresponds to the initial monomial of f under the term order obtained by refining the total degree partial order with the term order $>$.

A *multiplicative function* is an element $\alpha \in \Gamma$ such that $\alpha(1) = 1$ and $\alpha(ab) = \alpha(a)\alpha(b)$ whenever a and b are relatively prime. Cashwell and Everett observes that a multiplicative function is necessarily a unit in Γ . One can further observe that if α is multiplicative, then $f = \Phi(\alpha)$ can be written

$$f(x_1, x_2, x_3, \dots) = f_1(x_1)f_2(x_2)f_3(x_3)\cdots$$

where each $f_i(x_i) \in K[[x_i]]$ is invertible. In particular, the constant function $\Gamma \ni \nu_0$ with $\nu_0(n) = 1$ for all n , corresponds to

$$\sum_{m \in \mathcal{M}} m = \frac{1}{1-x_1} \frac{1}{1-x_2} \frac{1}{1-x_3} \cdots$$

Since the Möbius function is defined to be the inverse of this function, we get that it corresponds to

$$(1-x_1)(1-x_2)(1-x_3)\cdots = 1 - \left(\sum_{i=1}^{\infty} x_i\right) + \left(\sum_{i<j} x_i x_j\right) - \left(\sum_{i<j<k} x_i x_j x_k\right) + \cdots$$

2.2. Truncations of the ring of number-theoretic functions. Let $n, n' \in \mathbb{N}^+$, $n' > n$. Then there is a K -algebra epimorphism

$$\begin{aligned} \varphi_n^{n'} : \Gamma_{n'} &\rightarrow \Gamma_n \\ \varphi_n^{n'}(f)(m) &= \begin{cases} f(m) & m \leq n \\ 0 & m > n \end{cases} \end{aligned}$$

Hence, the Γ_n 's form an inverse system.

Lemma 2.1. $\varprojlim \Gamma_n \simeq \Gamma$.

Proof. Given any $f \in \Gamma$, the sequence $(\pi_1(f), \pi_2(f), \pi_3(f), \dots)$ is coherent. Conversely, given any coherent sequence $(g_1, g_2, g_3, \dots)$, we can define $g : \mathbb{N} \rightarrow K$ by $g(m) = g_i(m)$ where $i \geq m$. $\square$

As a side remark, we note that

Lemma 2.2. *The decreasing filtration*

$$J_1 \supsetneq J_2 \supsetneq J_3 \supsetneq \cdots \tag{11}$$

is separated, that is, $\cap_n J_n = (0)$.

Definition 2.3. We define

$$I_n = K[[X]] \{ m \in \mathcal{M} \mid w(m) > n \}, \tag{12}$$

that is, as the monomial ideal in $K[[X]]$ generated by all monomials of weight strictly higher than n . We put $A_n = \frac{K[[X]]}{I_n}$.

Proposition 2.4. *A K -basis of A_n is given by all monomials of weight $\leq n$. Hence A_n is an artinian algebra, with $\dim_K(A_n) = n$. Putting $r = r(n)$, we have that*

$$A_n = \frac{K[[X]]}{I_n} \simeq \frac{K[x_1, \dots, x_r]}{I_n \cap K[x_1, \dots, x_r]} \quad (13)$$

Proof. As a vector space, $K[[X]] \simeq U \oplus I_n$, where U consists of all functions supported on monomials of weight $\leq n$. It follows that $A_n \simeq U$ as K vector spaces. Of course, there are exactly n monomials of weight $\leq n$. Finally, if $s > r$ then $w(x_s) = p_s > n$, hence $x_s \in I_n$. $\square$

We will abuse notations and identify I_n and its contraction $I_n \cap K[x_1, \dots, x_r]$.

Lemma 2.5. $\Gamma_n \simeq A_n$.

Proof. Since A_n has a K -basis is given by all monomials of weight $\leq n$, the two K -algebras are isomorphic as K -vector spaces. The multiplication in A_n is induced from the multiplication in $K[[X]]$, with the extra condition that monomials of weight $> n$ are truncated. This is the same multiplication as in Γ_n . $\square$

Proposition 2.6. I_n is a strongly stable ideal, with respect to the reverse order of the variables.

Proof. We must show that if $m \in I_n$, and $x_i | m$, then $mx_j/x_i \in I$ for $i \leq j \leq r$. We have that $w(mx_j/x_i) = w(m)p_j/p_i > w(m) > n$. $\square$

Part ?? of the main theorem is now proved.

We give $K[x_1, \dots, x_r]$ an $\mathbb{N}^2$ -grading by giving the variable x_i bi-degree $(1, p_i)$. Since each I_n is bihomogeneous, this grading is inherited by A_n .

Theorem 2.7. *The bi-graded Hilbert series of A_n is given by*

$$A_n(t, u) = \sum_{i,j} c_{ij} t^i u^j,$$

where c_{ij} is the number of $p_1^{a_1} \dots p_r^{a_r} \leq n$ with $\sum a_r = i$ and $\sum a_r p_r = j$. Furthermore,

$$A_n(t, 1) = \sum_i d_i t^i$$

$$A_n(1, u) = \sum_j e_j u^j$$

where d_i is the number of $w \leq n$ with $\lambda(w) = i$, and e_i is the number of $w \leq n$ with $\tilde{\lambda}(w) = i$. In particular, the t^1 -coefficient of $A_n(t, 1)$ is the number of prime numbers $\leq n$.

Proof. The monomial $x_1^{a_1} \dots x_n^{a_n}$ has bi-degree $(\sum_{i=1}^n a_i, \sum a_i p_i)$. $\square$

This establishes part ?? of the main theorem.

3. MINIMAL GENERATORS FOR I_n

Let $n \in \mathbb{N}^+$, and let $r = r(n)$. We have that

$$x_1^{a_1} \dots x_r^{a_r} = m \in I_n \iff w(m) > n \iff \prod_{i=1}^r p_i^{a_i} > n. \quad (14)$$

We denote by $G(I_n)$ the set of minimal monomial generators of I_n . For $m = x_1^{a_1} \dots x_r^{a_r}$ to be an element of $G(I_n)$ it is necessary and sufficient that $m \in I_n$ and that for $1 \leq v \leq r$, $x_v | m \implies m/x_v \notin I_n$. In other words,

$$1 \leq j \leq n, a_j > 0 \implies n < \prod_{i=1}^r p_i^{a_i} \leq p_j n. \quad (15)$$

Definition 3.1. For n, v, d positive integers, we define:

$$C_n = \#G(I_n) \quad (16)$$

$$C_{n,v} = \# \{ m \in G(I_n) \mid \min(m) = v \} \quad (17)$$

$$C_{n,v,d} = \# \{ m \in G(I_n) \mid \min(m) = v, |m| = d \} \quad (18)$$

Theorem 3.2. $C_{n,v}$ is the number of solutions $(b_1, \dots, b_r) \in \mathbb{N}^r$ to the equation

$$\log n - \log p_v < \sum_{i=v}^r b_i \log p_i \leq \log n. \quad (19)$$

Equivalently, $C_{n,v}$ is the number of integers x such that $n/p_v < x \leq n$ and such that no prime factors of x are smaller than p_v .

Similarly, $C_{n,v,d}$ is the number of solutions $(b_1, \dots, b_r) \in \mathbb{N}^r$ to the system of equations

$$\begin{aligned} \log n - \log p_v < \sum_{i=v}^r b_i \log p_i \leq \log n \\ \sum_{i=1}^r b_i = d - 1. \end{aligned} \quad (20)$$

or equivalently, $C_{n,v,d}$ is the number of integers x such that $n/p_v < x \leq n$ and such that no prime factors of x are smaller than p_v , and with the additional constraint that $\lambda(x) = d$.

Proof. We have that $a_v > 0$, $a_w = 0$ for $w < v$. Hence equation (??) implies that

$$n < \prod_{j=v}^r p_j^{a_j} \leq p_v n.$$

Putting $b_v = a_v - 1$, $b_j = a_j$ for $j > v$ we can write this as

$$n < p_v \prod_{j=v}^r p_j^{b_j} \leq p_v n \iff n/p_v < \prod_{j=v}^r p_j^{b_j} \leq n$$

from which (??) follows by taking logarithms. This implies (??) as well. $\square$

We have now proved part ?? of the main theorem.

Example 3.3. The first few I_n 's are as follows: $I_2 = (x_1^2)$, $I_3 = (x_1^2, x_2^2, x_1 x_2)$, $I_4 = (x_1^3, x_2^2, x_1 x_2)$, $I_5 = (x_1^3, x_2^2, x_1 x_2, x_3^2, x_1 x_3, x_2 x_3)$.

We tabulate $C_{n,i}$ and $C_{n,i,d}$, the latter in form of the polynomial $u^{-2} \sum_j C_{n,i,j} u^j$ in the tables ?? and ??.

Theorem 3.4. (1) $C_{n,v} = 0$ for $v > r(n)$
 (2) $\forall n \in \mathbb{N} : \forall v \leq r(n) : C_{n,1+r(n)-v} \geq v$,
 (3) $\forall n \in \mathbb{N} : C_n \geq \binom{r(n)+1}{2}$,
 (4) $\forall v \in \mathbb{N} : \exists N : \forall n \geq N : C_{n,1+r(n)-v} = v$.
 (5) If n is even, then $C_{n,v} = C_{n-1,v}$ for all v ,
 (6) $C_{n,1} = \lceil n/2 \rceil$.

Proof. (??) Obvious.

(??) and (??) It suffices to show that for any subset $S \subset \{1, \dots, r\}$ of cardinality 1 or 2, there is an $m \in G(I_n)$ with $\text{Supp}(m) = S$. If $S = \{i\}$ then there is a unique positive integer a such that $p_i^{b-1} \leq n < p_i^b$, and $m = x_i^a$ is the desired generator. If $S = \{i, j\}$ with $i < j$ then we claim that there is a positive integer a such that $x_i^a x_j \in G(I_n)$. Namely, choose b such that $p_i^{b-1} \leq n < p_i^b$, then since $p_i < p_j$ one has $n < p_i^{b-1} p_j$. Hence $x_i^{b-1} x_j \in I_n$, so it is a multiple of some minimal generator. By the definition of b , this minimal generator must be of the form $x_i^a x_j$ for some a , which establishes the claim.

(??) We must show that the number of solutions in $\mathbb{N}^r$ to

$$\frac{n}{2} < \prod_{i=1}^r p_i^{b_i} \leq n$$

is precisely $\lceil \frac{n}{2} \rceil$. Obviously, any integer $\in (\frac{n}{2}, n]$ fits the bill; there are $\lceil \frac{n}{2} \rceil$ of those.

(??) The case $v = 1$ follows from (??). Hence, it suffices to show that if $v > 1$, $x \in (\frac{n}{p_v}, n] \cap \mathbb{N}$, and if x has no prime factor $< p_v$, then $x \in (\frac{n-1}{p_v}, n-1] \cap \mathbb{N}$. The only way this can fail to happen is if $x = n$, but then x is even, and has the prime factor $2 = p_1 < p_v$, a contradiction.

(??) For large enough n , the only integers $x \leq n$ with all prime factors $\geq 1 + r(n) - v$ are $p_{1+r(n)-v}, \dots, p_{r(n)}$. There is v of these, and they are all $> \frac{n}{p_v}$. $\square$

Theorem 3.5. (1) $C_{n,v,d} = 0$ for $v > r(n)$, and for $d < 2$,
 (2) $\forall v \in \mathbb{N} : \exists N : \forall n \geq N : C_{n,1+r(n)-v,2} = v$, $C_{n,1+r(n)-v,d} = 0$ for $d \neq 2$,
 (3) $\binom{r(n)}{2} = \# \{ m \in \mathbb{N}^+ | m \leq n, \lambda(m) = 2 \}$.

Proof. The first and the last assertions are obvious. The second one follows from the proof of (??) in the previous lemma. $\square$

4. POINCARÉ SERIES

In [?], a minimal free multi-graded resolution of a I over S is given, where $S = K[x_1, \dots, x_r]$ is a polynomial ring, and $I \subset (x_1, \dots, x_r)^2$ is a stable ideal. As a consequence, the following formula for the Poincaré-Betti series is derived:

$$P(\text{Tor}_*^S(I, K), t) = \sum_{a \in G(I)} (1+t)^{\max(a)-1} \quad (21)$$

where $G(I)$ is the minimal generating set of I . Since the resolution is multi-graded, (??) can be modified to yield a formula for the graded Poincaré-Betti series (we here consider S as $\mathbb{N}$ -graded, with each variable given weight 1):

$$P(\text{Tor}_{*,*}^S(I, K), t, u) = \sum_{a \in G(I)} u^{|a|} (1+t)^{\max(a)-1} \quad (22)$$

We will use the following variant of this result:

Theorem 4.1 (Eliashou-Kervaire). *Let $I \subset (x_1, \dots, x_r)^2 \subset K[x_1, \dots, x_r] = S$ be a stable monomial ideal. Put*

$$b_{i,d} = \# \{ m \in G(I) | \max(m) = i, |m| = d \} \quad (23)$$

$$b_i = \# \{ m \in G(I) | \max(m) = i \} \quad (24)$$

Then

$$P(\text{Tor}_*^S(I, K), t) = \sum_{i=1}^r b_i (1+t)^{(i-1)} \quad (25)$$

$$P(\text{Tor}_{*,*}^S(I, K), t, u) = \sum_{i=1}^r \left((1+tu)^{(i-1)} \sum_j b_{i,j} u^j \right). \quad (26)$$

For the Betti-numbers we have that

$$\beta_q = \dim_K (\text{Tor}_q^S(I, K)) = \sum_{i=1}^r b_i \binom{i-1}{q}. \quad (27)$$

From Proposition ?? we have that the ideals I_n are stable after reversing the order of the variables. Hence, replacing $\max$ by $\min$, and hence b_i with $C_{n,1+r-i}$, we get:

Corollary 4.2. *Let $n \in \mathbb{N}^+$, $r = r(n)$, $S = K[x_1, \dots, x_r]$. Then*

$$P(\mathrm{Tor}_*^S(I_n, K), t) = \sum_{i=1}^r C_{n,1+r-i} (1+t)^{(i-1)} \quad (28)$$

$$P(\mathrm{Tor}_{*,*}^S(I_n, K), t, u) = \sum_{i=1}^r (1+tu)^{(i-1)} \sum_j C_{n,1+r-i,j} u^j. \quad (29)$$

For the Betti-numbers we have that

$$\beta_q = \sum_{i=1}^r C_{n,1+r-i} \binom{i-1}{q}. \quad (30)$$

In [?, ?] it is shown that if $S = K[x_1, \dots, x_r]$ and I is a stable monomial ideal in S , then S/I is a Golod ring. Hence, from a result of Golod [?] (see also [?]), it follows that

$$P(\mathrm{Tor}_*^{S/I}(K, K), t) = \frac{(1+t)^r}{1 - t^2 P(\mathrm{Tor}_*^S(I, K), t)} \quad (31)$$

Regarding S as an $\mathbb{N}$ -graded ring, one can show that in fact

$$P(\mathrm{Tor}_*^{S/I}(K, K), t, u) = \frac{(1+ut)^r}{1 - t^2 P(\mathrm{Tor}_*^S(I, K), t, u)} \quad (32)$$

The following theorem is an immediate consequence:

Theorem 4.3 (Herzog-Aramova, Peeva). *Let $S = K[x_1, \dots, x_r]$, and suppose that I is a stable monomial ideal in S . Put*

$$\begin{aligned} b_{i,d} &= \# \{ x \in G(I) \mid \max(x) = i, |x| = d \} \\ b_i &= \# \{ x \in G(I) \mid \max(x) = i \} \end{aligned}$$

Then, for $R = S/I$, we have that

$$P(\mathrm{Tor}_*^R(K, K), t) = \frac{(1+t)^r}{1 - t^2 \sum_{i=1}^r (1+t)^{(i-1)} \sum_j b_i} \quad (33)$$

$$P(\mathrm{Tor}_*^R(K, K), t, u) = \frac{(1+t)^r}{1 - t^2 \sum_{i=1}^r (1+tu)^{(i-1)} \sum_j b_{i,j} u^j} \quad (34)$$

Specialising to the case of A_n , we obtain:

Corollary 4.4. *Let $n \in \mathbb{N}^+$, and let $r = r(n)$. Regard A_n as a naturally graded K -algebra, with each x_i given weight 1, and regard K as a cyclic A -module. Then*

$$P(\mathrm{Tor}_*^{A_n}(K, K), t) = \frac{(1+t)^r}{1 - t^2 \sum_{i=1}^r (1+t)^{(i-1)} C_{n,r-i+1}} \quad (35)$$

$$P(\mathrm{Tor}_*^{A_n}(K, K), t, u) = \frac{(1+ut)^r}{1 - t^2 \left(\sum_{i=1}^r \left((1+tu)^{(i-1)} \sum_j C_{n,r-i+1,j} u^j \right) \right)} \quad (36)$$

Part ?? of the main theorem is now proved.

Example 4.5. We consider the case $n = 5$, then $r = r(n) = 3$, so $S = K[x_1, x_2, x_3]$ and $I = I_5 = (x_1^3, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2)$. We get that $C_{5,1} = 3, C_{5,2} = 2, C_{5,3} = 1$. According to our formulas¹ we have

$$\begin{aligned} P_I^S(t) &= 1 + 2(1+t) + 3(1+t)^2 = 6 + 8t + 3t^2 \\ P_K^{S/I} &= \frac{(1+t)^r}{1 - t^2 P_I^S(t)} = \frac{1}{1 - 3t} \end{aligned}$$

¹Here, we have used the abbreviation $P_I^S(t) = P(\mathrm{Tor}_*^S(I, K), t)$, we will also write $P_K^{S/I}(t) = P(\mathrm{Tor}_*^{S/I}(K, K), t)$ et cetera.

When we consider the grading by total degree, we have that $C_{5,1,2} = 2$, $C_{5,1,3} = 1$, $C_{5,2,2} = 2$, $C_{5,3,2} = 1$. Hence, our formulas yield

$$\begin{aligned} P_I^S(t, u) &= u^2 + 2u^2(1+t) + (2u^2 + u^3)(1+t)^2 \\ &= 5u^2 + u^3 + (6u^2 + 2u^3)t + (2u^2 + u^3)t^2 \\ P_K^{S/I}(t, u) &= -\frac{1+tu}{u^3t^2 + 2t^2u^2 + 2tu - 1} \end{aligned}$$

We list the first few Poincaré-Betti series $P(\mathrm{Tor}_*^{A_n}(K, K), t, u)$ in table ??.

Conjecture 4.6. $P(\mathrm{Tor}_*^{A_n}(K, K), t) = -\frac{(1+t)^{\ell_1(n)}}{q_n(t)}$, $q_n(t) = \sum_{i=0}^{\ell_2(n)} h_i(n)t^i$, with

- (1) $q_n(-1) \neq 0$,
- (2) $\ell_1(n)$ is the number of odd primes p such that $p^2 \leq n$,
- (3) $\ell_2(n) = \ell_1(n) + 1$,
- (4) $h_0(n) = -1$,
- (5) $h_1(n) = r(n) - \ell_1(n)$,
- (6) $h_{\ell_2(n)}(n) = C_{n,1} = \lceil n/2 \rceil$.

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REFERENCES

- [1] Annetta Aramova and Jürgen Herzog. Koszul Cycles and Eliahou-Kervaire Type Resolutions. *Journal of Algebra*, 181(2):347–370, 1996.
- [2] E. D. Cashwell and C. J. Everett. The ring of number-theoretic functions. *Pacific Journal of Mathematics*, 9:975–985, 1959.
- [3] S. Eliahou and M. Kervaire. Minimal resolutions of some monomial ideals. *J. Algebra*, 129:1–25, 1990.
- [4] E. S. Golod. On the homology of some local rings. *Soviet Math. Dokl.*, 3:745–749, 1962.
- [5] T. Gulliksen and G. Levin. *Homology of Local Rings*, volume 20 of *Queen's Papers in Pure and Applied Mathematics*. 1969.
- [6] Irena Peeva. 0-Borel Fixed Ideals. *Journal of Algebra*, 184(3):945–984, 1996.

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FIGURE 1. The numbers C_n and $C_{n,i}$.

n	Σ	$i = 1$	$i = 2$	3	4	5	6	7	8	9	10
2	1	1									
3	3	2	1								
4	3	2	1								
5	6	3	2	1							
6	6	3	2	1							
7	10	4	3	2	1						
8	10	4	3	2	1						
9	11	5	3	2	1						
10	11	5	3	2	1						
11	16	6	4	3	2	1					
12	16	6	4	3	2	1					
13	22	7	5	4	3	2	1				
14	22	7	5	4	3	2	1				
15	23	8	5	4	3	2	1				
16	23	8	5	4	3	2	1				
17	30	9	6	5	4	3	2	1			
18	30	9	6	5	4	3	2	1			
19	38	10	7	6	5	4	3	2	1		
20	38	10	7	6	5	4	3	2	1		
21	39	11	7	6	5	4	3	2	1		
22	39	11	7	6	5	4	3	2	1		
23	48	12	8	7	6	5	4	3	2	1	
24	48	12	8	7	6	5	4	3	2	1	
25	50	13	9	7	6	5	4	3	2	1	
26	50	13	9	7	6	5	4	3	2	1	
27	51	14	9	7	6	5	4	3	2	1	
28	51	14	9	7	6	5	4	3	2	1	
29	61	15	10	8	7	6	5	4	3	2	1
30	61	15	10	8	7	6	5	4	3	2	1

FIGURE 2. The numbers $C_{n,i,g}$.

n	$i = 1$	$i = 2$	3	4	5	6	7	8	9	
2	1									
3	2	1								
4	$u + 1$	1								
5	$u + 2$	2	1							
6	$2u + 1$	2	1							
7	$2u + 2$	3	2	1						
8	$u^2 + u + 2$	3	2	1						
9	$u^2 + 2u + 2$	$u + 2$	2	1						
10	$u^2 + 3u + 1$	$u + 2$	2	1						
11	$u^2 + 3u + 2$	$u + 3$	3	2	1					
12	$2u^2 + 2u + 2$	$u + 3$	3	2	1					
13	$2u^2 + 2u + 3$	$u + 4$	4	3	2	1				
14	$2u^2 + 3u + 2$	$u + 4$	4	3	2	1				
15	$2u^2 + 4u + 2$	$2u + 3$	4	3	2	1				
16	$u^3 + u^2 + 4u + 2$	$2u + 3$	4	3	2	1				
17	$u^3 + u^2 + 4u + 3$	$2u + 4$	5	4	3	2	1			
18	$u^3 + 2u^2 + 3u + 3$	$2u + 4$	5	4	3	2	1			
19	$u^3 + 2u^2 + 3u + 4$	$2u + 5$	6	5	4	3	2	1		
20	$u^3 + 3u^2 + 2u + 4$	$2u + 5$	6	5	4	3	2	1		
21	$u^3 + 3u^2 + 3u + 4$	$3u + 4$	6	5	4	3	2	1		
22	$u^3 + 3u^2 + 4u + 3$	$3u + 4$	6	5	4	3	2	1		
23	$u^3 + 3u^2 + 4u + 4$	$3u + 5$	7	6	5	4	3	2	1	
24	$2u^3 + 2u^2 + 4u + 4$	$3u + 5$	7	6	5	4	3	2	1	
25	$2u^3 + 2u^2 + 5u + 4$	$4u + 5$	$u + 6$	6	5	4	3	2	1	
26	$2u^3 + 2u^2 + 6u + 3$	$4u + 5$	$u + 6$	6	5	4	3	2	1	
27	$2u^3 + 3u^2 + 6u + 3$	$u^2 + 3u + 5$	$u + 6$	6	5	4	3	2	1	
28	$2u^3 + 4u^2 + 5u + 3$	$u^2 + 3u + 5$	$u + 6$	6	5	4	3	2	1	
29	$2u^3 + 4u^2 + 5u + 4$	$u^2 + 3u + 6$	$u + 7$	7	6	5	4	3	2	1
30	$2u^3 + 5u^2 + 4u + 4$	$u^2 + 3u + 6$	$u + 7$	7	6	5	4	3	2	1

n	<i>Graded</i>	<i>Non – graded</i>
2	$-(tu-1)^{-1}$	$-(t-1)^{-1}$
3	$-(2tu-1)^{-1}$	$-(2t-1)^{-1}$
4	$-\frac{1+tu}{(u^3+u^2)t^2+tu-1}$	$-(2t-1)^{-1}$
5	$-\frac{1+tu}{(u^3+2u^2)t^2+2tu-1}$	$-(3t-1)^{-1}$
6	$-\frac{1+tu}{(2u^3+u^2)t^2+2tu-1}$	$-(3t-1)^{-1}$
7	$-\frac{1+tu}{(2u^3+2u^2)t^2+3tu-1}$	$-(4t-1)^{-1}$
8	$-\frac{1+tu}{(u^4+u^3+2u^2)t^2+3tu-1}$	$-(4t-1)^{-1}$
9	$-\frac{1+2tu+t^2u^2}{(u^5+2u^4+2u^3)t^3+(u^4+3u^3+4u^2)t^2+2tu-1}$	$-\frac{1+t}{5t^2+3t-1}$
10	$-\frac{1+2tu+t^2u^2}{(u^5+3u^4+u^3)t^3+(u^4+4u^3+3u^2)t^2+2tu-1}$	$-\frac{1+t}{5t^2+3t-1}$
11	$-\frac{1+2tu+t^2u^2}{(u^5+3u^4+2u^3)t^3+(u^4+4u^3+5u^2)t^2+3tu-1}$	$-\frac{1+t}{6t^2+4t-1}$
12	$-\frac{1+2tu+t^2u^2}{(2u^5+2u^4+2u^3)t^3+(2u^4+3u^3+5u^2)t^2+3tu-1}$	$-\frac{1+t}{6t^2+4t-1}$
13	$-\frac{1+2tu+t^2u^2}{(2u^5+2u^4+3u^3)t^3+(2u^4+3u^3+7u^2)t^2+4tu-1}$	$-\frac{1+t}{7t^2+5t-1}$
14	$-\frac{1+2tu+t^2u^2}{(2u^5+3u^4+2u^3)t^3+(2u^4+4u^3+6u^2)t^2+4tu-1}$	$-\frac{1+t}{7t^2+5t-1}$
15	$-\frac{1+2tu+t^2u^2}{(2u^5+4u^4+2u^3)t^3+(2u^4+6u^3+5u^2)t^2+4tu-1}$	$-\frac{1+t}{8t^2+5t-1}$
16	$-\frac{1+2tu+t^2u^2}{(u^6+u^5+4u^4+2u^3)t^3+(u^5+u^4+6u^3+5u^2)t^2+4tu-1}$	$-\frac{1+t}{8t^2+5t-1}$
17	$-\frac{1+2tu+t^2u^2}{(u^6+u^5+4u^4+3u^3)t^3+(u^5+u^4+6u^3+7u^2)t^2+5tu-1}$	$-\frac{1+t}{9t^2+6t-1}$
18	$-\frac{1+2tu+t^2u^2}{(u^6+2u^5+3u^4+3u^3)t^3+(u^5+2u^4+5u^3+7u^2)t^2+5tu-1}$	$-\frac{1+t}{9t^2+6t-1}$
19	$-\frac{1+2tu+t^2u^2}{(u^6+2u^5+3u^4+4u^3)t^3+(u^5+2u^4+5u^3+9u^2)t^2+6tu-1}$	$-\frac{1+t}{10t^2+7t-1}$
20	$-\frac{1+2tu+t^2u^2}{(u^6+3u^5+2u^4+4u^3)t^3+(u^5+3u^4+4u^3+9u^2)t^2+6tu-1}$	$-\frac{1+t}{10t^2+7t-1}$
21	$-\frac{(1+tu)^2}{t^3u^6+3u^5t^3+t^2u^5+3t^3u^4+3u^4t^2+4t^3u^3+6t^2u^3+8t^2u^2+6tu-1}$	$-\frac{1+t}{11t^2+7t-1}$
22	$-\frac{(1+tu)^2}{t^3u^6+3u^5t^3+t^2u^5+4t^3u^4+3u^4t^2+3t^3u^3+7t^2u^3+7t^2u^2+6tu-1}$	$-\frac{1+t}{11t^2+7t-1}$
23	$-\frac{(1+tu)^2}{t^3u^6+3u^5t^3+t^2u^5+4t^3u^4+3u^4t^2+4t^3u^3+7t^2u^3+9t^2u^2+7tu-1}$	$-\frac{1+t}{12t^2+8t-1}$
24	$-\frac{(1+tu)^2}{2t^3u^6+2u^5t^3+2t^2u^5+4t^3u^4+2u^4t^2+4t^3u^3+7t^2u^3+9t^2u^2+7tu-1}$	$-\frac{1+t}{12t^2+8t-1}$
25	$-\frac{(1+tu)^3}{q(t,u)}$	$-\frac{(1+t)^2}{13t^3+22t^2+7t-1}$

FIGURE 3. Graded and non-graded Poincaré-Betti series of the minimal free resolution of K over A_n .